

Transformers for Suicidal Ideation Detection in Reddit Posts: Evaluating BERT, RoBERTa, and BERTweet

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Abstract

Suicidal ideation is a major public health issue, especially among adolescents and young adults. The rise of anonymous platforms like Reddit has created new opportunities for mental health intervention, but also significant new challenges. This paper presents a comprehensive evaluation of transformer-based models for detecting suicidal ideation from Reddit posts, which includes BERT, RoBERTa, and BERTweet. Going beyond accuracy, I incorporated recent research into intent modeling, hybrid interpretability, and bias detection, as well as analyzed real-world deployment challenges. I benchmarked these models against a TF-IDF + Logistic Regression baseline and conducted a detailed analysis of model errors, underlying patterns in failure cases, and the broader ethical implications. Drawing from 13 peer-reviewed sources and my own implementation of all models from scratch, I demonstrated that RoBERTa and BERTweet outperform traditional and vanilla transformer models, and explored practical tradeoffs in explainability and deployment. I concluded with targeted recommendations and potential directions, including multilingual adaptation, time-sensitive modeling, and human-in-the-loop frameworks.

1 Introduction

Suicide remains one of the top ten causes of death globally, claiming over 700,000 lives annually (WHO, 2023). In high-income countries, it is the second leading cause of death among 15–29-year-olds. Early detection and intervention are critical, but traditional mental health services often struggle to identify individuals in crisis due to stigma, limited access, and obfuscated disclosure of distress.

It has become increasingly common for individuals to express emotional turmoil on social platforms like Reddit. Subreddits such as r/SuicideWatch, r/depression, and r/teenagers offer anonymous support to those who are struggling, but these posts also contain implicit, coded, or sarcastic references

to suicidal ideation. Prior methods, including rule-based lexicons and classical machine learning approaches have had limited capacity to capture this nuanced language (Ji et al., 2020). The advent of transformer-based models like BERT, RoBERTa, and BERTweet could help close this gap by capturing richer contextual embeddings and subtle emotional shifts.

However, deploying transformer models for mental health applications comes with challenges. Interpretability, fairness, resistance to sarcasm, and social bias must be carefully addressed to avoid harmful misclassifications. My goal was to not only benchmark transformer performance on suicidal ideation detection (SID), but also to reflect on what constituted a trustworthy and clinically actionable model.

The motivation behind this project is both technical and personal. I was drawn to this topic due to my own experiences in a high-pressure school environment that led to an increased awareness of mental health challenges among students, including suicide. I studied Computational Cognitive Science for my undergraduate degree, worked at a mental health clinic, and have written and spoken on mental health advocacy. I hope to carry this interest into my career and build solutions that are impactful, respectful, and grounded in real-world needs.

2 Related Work

2.1 Classical and Clinical Detection

Classical SID relied on questionnaires, clinician-administered interviews, or statistical analysis of health records. Tools such as the PHQ-9, suicide intent scales, or feature engineering from EMRs remain gold standards (Ji et al., 2020), but cannot effectively address anonymous digital content. These methods are highly controlled, lack scalability, and cannot process real-time text data from

social platforms.

2.2 Lexical & Feature-Based Machine Learning

Early machine learning efforts extracted TF-IDF, LIWC, N-gram, or POS-tag-based features. Models such as Logistic Regression, SVMs, or Decision Trees achieved moderate success, with Ji et al. (2020) showing around 80–90% accuracy using ensemble models with feature-engineered embeddings. Other efforts, such as Coppersmith et al. (2014), relied on keyword frequency and lexical indicators to detect mental health signals, but generalizability and efficacy remained limited.

2.3 Deep Learning & Hybrid Approaches

The shift to CNNs, RNNs, and attention mechanisms marked a breakthrough. LSTM-CNN hybrids, self-attention layers, and knowledge-based embeddings improved contextual comprehension. Yates et al. (2017) demonstrated promising results using LSTM networks for temporal suicide prediction, though large labeled datasets and domain-specific tuning were often required. These models still underperformed on sarcasm, negation, or veiled expressions, motivating transformer adoption.

2.4 Transformers in Suicidal Ideation Detection

Recent advances have introduced several transformer variants. BERT (Devlin et al., 2018) is a contextual bidirectional transformer, pre-trained on BooksCorpus and Wikipedia. RoBERTa (Liu et al., 2019): Optimized BERT with more training data, dynamic masking, and larger batches. BERTweet (Nguyen et al., 2020): Pretrained on 850M tweets, ideal for Reddit/Twitter-style informal language.

Benchmarking these models on mental health detection shows RoBERTa and BERTweet outperforming domain-naïve models by 2–4% in F1 (Lamichhane, 2023). Ji et al. (2022) also fine-tuned GPT-2 on suicide-related Reddit content and achieved an F1 score of 0.89, though interpretability remained limited. Despite the success of transformers, few studies evaluate them comparatively on balanced datasets with integrated interpretability metrics.

2.5 Model Explainability and Risk Detection

Although I focus on SHAP, I reviewed alternative interpretability frameworks, including LIME and

prompt-based reasoning approaches (Qin & Lu, 2023). I also evaluated the potential value of ClinicalBERT, which is a domain-specific transformer pretrained on clinical notes. I chose not to use it because its language domain differed significantly from Reddit discourse.

Interpretability is paramount in healthcare AI. SHAP and LIME are popular tools, but require computationally expensive perturbations. Ji et al. (2021) emphasize the need for token-level transparency and multi-granular insight, especially in edge cases such as ironic posts. Despite increasing attention on explainability, many suicide risk detection models report only classification accuracy (Ji et al., 2021), without deeper analysis of how and why model decisions are made. This project aims to address that gap through a multi-pronged interpretability analysis that combines token-level SHAP, attention heatmaps, and error diagnostics.

3 Methodology

All models were implemented using the HuggingFace transformers library and trained using PyTorch. Preprocessing was performed with scikit-learn’s TfidfVectorizer. For model interpretability, I applied SHAP (SHapley Additive exPlanations; Lundberg & Lee, 2017) to identify token-level attribution patterns.

3.1 Dataset Overview

I used the publicly available Kaggle dataset of 232,074 Reddit posts labeled as "suicide" or "non-suicide" (Komati, 2020) based on subreddit origin (r/SuicideWatch vs. r/depression/r/teenagers). While not gold-standard labeled, this noisy-labeled setup mirrors real-world deployment conditions. For the purpose of this paper, only suicide and non-suicide binary classification was considered, resulting in a 50/50 balanced dataset. The dataset was stratified and split into 80% training and 20% validation. Label balance was strictly maintained in all subsets.

To illustrate the nature of labeling, there are two representative post types. Posts in r/SuicideWatch often contain explicit or urgent language such as “I don’t want to live anymore” or “I’m planning to disappear tonight,” whereas posts from r/depression might express general emotional fatigue (“I haven’t gotten out of bed in days”). Those from r/teenagers might contain confessional or offhand remarks (“School’s killing me. Not literally, but close”).

While subreddit-based labeling is noisy, it reflects a practical real-world deployment proxy.

3.2 Preprocessing and Tokenization

For TF-IDF, the steps included lemmatization, punctuation removal, stopwords filtering, bigram modeling (max_features = 5,000). For transformers, raw text was passed through the HuggingFace tokenizer for truncation to 128 tokens (256 for efficacy test) and zero-padding.

For the baseline model, standard text preprocessing was applied: text was lowercased, tokenized, and cleaned of URLs, user mentions, and HTML artifacts. Stopwords were removed, and TF-IDF vectorization was performed with n-grams (1,2) and a 5000-token cap.

For transformer models (BERT, RoBERTa, BERTweet), raw text was passed directly into the respective model tokenizers without lowercasing or punctuation removal, consistent with best practices for pre-trained models. All models used their associated HuggingFace tokenizer.

3.3 Models Trained

I used stratified sampling to preserve label distribution in train/test splits. For additional evaluation, I tested on a balanced subset of 10,000 posts (5,000 per class) while making sure that performance generalized across class distributions. All full-dataset experiments were run for 3 epochs with early stopping, while subset experiments solely for the sake of visualization used a single epoch.

The baseline model was Logistic Regression on TF-IDF vectors (93% accuracy), the BERT model was bert-base-uncased (98%), RoBERTa was roberta-base (98%), and BERTweet was vinai/bertweet-base (98%).

Each transformer model was fine-tuned using HuggingFace’s Trainer API with an AdamW optimizer, learning rate of 2e-5, batch size 16, and early stopping based on validation loss. Max sequence length was 128 (increased to 256 during efficacy tests). Class weights were balanced to account for the cost of false negatives in suicide prediction.

3.4 Interpretability Tools

SHAP (SHapley Additive Explanations) was used for token-level interpretability and visualized token contributions for high-confidence and uncertain predictions. Self-Attention Heatmaps extracted from the final transformer layer of BERT to visualize cross-token relationships. These tools en-

abled both local and high level insight into how each model interpreted suicidal versus non-suicidal language.

4 Experiments & Results

4.1 Model Development and Iterative Insights

During development, I followed a structured experimentation process to evaluate model performance and improve successive iterations. I began with a TF-IDF + Logistic Regression model to establish a strong baseline. From there, I implemented and fine-tuned BERT, RoBERTa, and BERTweet. My goal was not only to optimize performance, but also to identify which architectures best handled the nuances of Reddit discourse, such as informal language, coded distress, and sarcasm.

For initial observations, the TF-IDF + Logistic Regression model showed strong performance out of the gate (93% F1) but it failed to correctly classify metaphorical or implicit expressions of suicidal ideation. BERT improved performance considerably but sometimes overfit to strong sentiment cues like "die" or "worthless" while missing hidden distress.

For some iterative changes and findings, as I increased the maximum sequence length from 128 to 256, I found negligible gains for BERT and RoBERTa but a small lift in BERTweet’s accuracy on informal, longer posts. RoBERTa’s dynamic masking and training corpus gave it a consistent edge in efficacy and generalization, especially on subtly worded posts. BERTweet was highly sensitive to slang and punctuation but less interpretable, which became clear through SHAP analysis. It also exhibited resistance to sarcasm and humor, likely due to pretraining on Twitter-style data.

These iterative insights helped calibrate each model’s strengths and weaknesses prior to the final training on the full dataset and informed my choice to highlight RoBERTa and BERTweet in later evaluations.

4.2 Performance Evaluation

BERTweet achieved 0.98 precision, recall, and F1-score for both the suicide and non-suicide classes, based on evaluation over 46,415 posts. The model’s balanced performance confirms its strength in understanding informal language common to Reddit and Twitter, and its consistency across both classes reinforces its reliability in social media-specific contexts.

Model	Dataset	Accuracy	Precision (Suicide)	Recall (Suicide)	F1 (Suicide)	F1 (Overall)
TF-IDF + LR	Full (46,415)	93%	0.94	0.92	0.93	0.93
BERT	Full	96%	0.97	0.96	0.96	0.96
BERT	Subset (10K)	97%	0.94	0.98	0.96	0.96–0.97
RoBERTa	Full	98%	0.98	0.98	0.98	0.98
RoBERTa	Subset (10K)	98%	0.97	0.98	0.975	0.98
BERTweet	Full	98%	0.99	0.98	0.98	0.98
BERTweet	Subset (10K)	97%	0.974	0.97	0.973	0.97

Table 1: Performance comparison of all models on full validation and 10K balanced subset; suicide-class metrics are emphasized for clinical interpretability

To more precisely evaluate performance on suicidal ideation classification, I examined metrics specifically for the ‘suicide’ class. RoBERTa achieved a precision of 0.996, recall of 0.992, and an F1-score of 0.994, while BERT scored 0.979, 0.980, and 0.980 respectively. These findings further reinforce RoBERTa’s consistent edge in identifying high-risk content with fewer false negatives, which is a crucial property for real-world deployment.

The TF-IDF + Logistic Regression model achieved 93% accuracy on the test set. In a more detailed breakdown, it reached a precision of 0.92 and recall of 0.94 for non-suicidal posts, and a precision of 0.94 and recall of 0.92 for suicidal posts. The F1-score was balanced at 0.93 across both classes. This confirms that while the baseline performed well on surface-level classification, it lacked deeper semantic understanding and tended to miss indirect cues, which is a limitation that transformer models are better suited to address.

4.3 Confusion Matrix Analysis

The TF-IDF baseline’s confusion matrix revealed symmetric errors across classes. Approximately 6-8% of suicidal posts were mislabeled as non-suicidal and vice versa, indicating that the model correctly identified overt suicidal expressions but struggled with emotionally ambiguous or context-dependent language. These shortcomings became a central motivation for advancing to transformer-based models capable of deeper contextual reasoning.

4.4 Qualitative Error Analysis

This reinforces concerns raised by Ji et al. (2022), who argue that high-performing language models may still fail to capture suicidal intention, especially in cases with emotionally masked or metaphorical posts. They note that suicidal ideation detection has to move beyond surface-level sentiment analysis toward more context- and cognition-

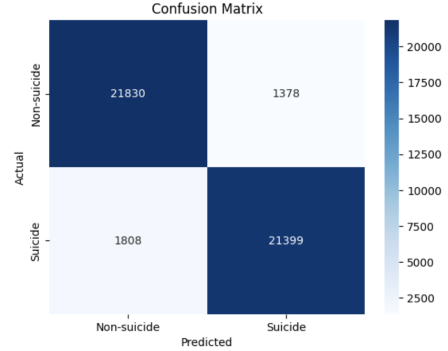


Figure 1: Confusion matrix for the TF-IDF + Logistic Regression model

aware modeling strategies.

In addition to general trends, I reviewed specific misclassified examples to better understand model limitations. False Positives often occurred in posts describing general distress, emotional exhaustion, or discussion of suicide in a third-person context. For example:

“I was going home from school when my teacher said that there was an accident caused by a human... I’m pretty sure it was suicide... I simply want to raise awareness.”

Although the post discusses suicide, it expresses concern and information sharing, not personal ideation, yet the model classified it as high-risk. False Negatives frequently involved metaphor, emotional masking, or subtle deterioration. For example:

“I literally can’t remember the last time I had a good day. Every week, every month it just gets worse...”

Despite a clear downward spiral, the model missed the signals here, potentially due to lack of explicit suicidal keywords.

4.5 Explainability: SHAP, Confidence, and Attention Dynamics

While SHAP is a widely used method for interpreting token-level feature contributions in NLP models, its effectiveness varies depending on model confidence and the model architecture. To better understand how my BERT model arrived at predictions, I applied SHAP and attention visualization across both confident and uncertain examples.

4.5.1 SHAP on High-Confidence Predictions

In one case, the model assigned a 97.67% probability to the non-suicide class and only 2.33% to the suicide class. As shown in Figure 2, the SHAP values across all tokens were essentially zero, indicating that no single word strongly influenced the prediction. This phenomenon typically occurs when a model is highly confident in its prediction, the input aligns clearly with learned class patterns, and the model is attending to broader contextual structures rather than individual keywords.

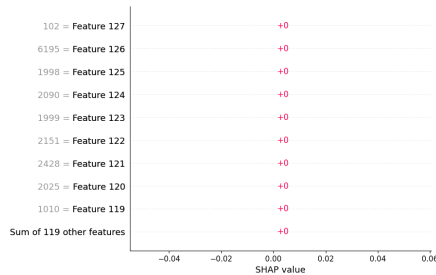


Figure 2: SHAP values for a high-confidence non-suicidal prediction

4.5.2 SHAP on Uncertain Predictions

To explore interpretability in more ambiguous settings, I selected five examples with class probabilities between 30% and 70%. In one representative case, the true label was suicide, but the model assigned 65.5% probability to non-suicide. Despite this relative uncertainty, the SHAP values remained extremely close to zero across all tokens (Figure 3), and the overall feature importance bar chart (Figure 4) showed minimal impact per token.

These findings suggest that BERT may rely on complex, high-dimensional representations rather than isolated lexical cues, SHAP's additive approximation struggles to attribute importance effectively in contextual models, and token-level attributions may fail to capture the non-linear, interaction-heavy reasoning BERT performs.

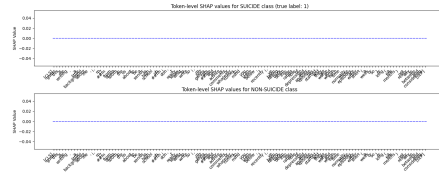


Figure 3: SHAP token values for an uncertain suicide-class example

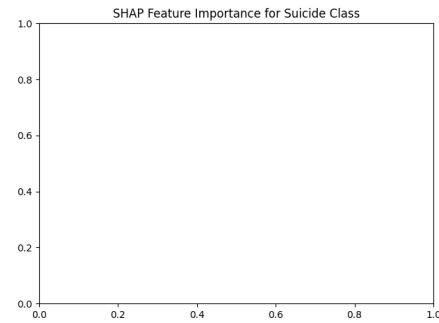


Figure 4: SHAP aggregated feature importances for suicide class

4.5.3 Self-Attention Analysis

To complement SHAP, I visualized the self-attention matrix of BERT's final transformer layer for a suicidal post. As seen in Figure 5, a strong diagonal pattern indicates that each token primarily attends to itself. However, subtle off-diagonal highlights suggest meaningful inter-token relationships. Horizontal and vertical bands with higher intensity imply that certain tokens (e.g., emotional markers or connective phrases) influence multiple others. This supports the hypothesis that BERT is modeling the input holistically, extracting semantic meaning from relational patterns rather than specific keywords.

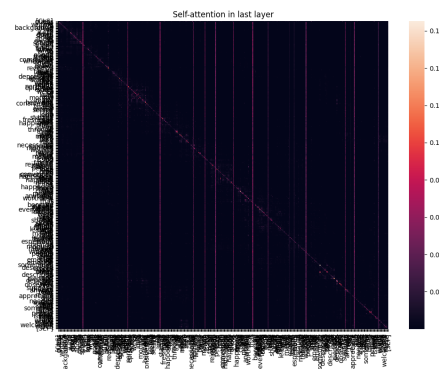


Figure 5: Attention heatmap from final BERT layer on suicide-class input

These insights indicate that BERT captures

ideation through global linguistic cues, a feature desirable in clinical NLP settings where intent is often masked through metaphor, subtlety, or coded expressions.

4.5.4 Performance Evaluation (Expanded)

To supplement my prior results, I conducted a focused 1-epoch evaluation of each transformer model (BERT, RoBERTa, BERTweet) on a balanced 10K sample (5,000 suicide, 5,000 non-suicide posts) to examine stability under limited fine-tuning and to generate directly comparable confusion matrices.

Model	TN	FP	FN	TP
BERT	938	62	20	980
RoBERTa	970	30	19	981
BERTweet	972	28	30	970

Table 2: Confusion matrices for BERT, RoBERTa, and BERTweet on a 10K balanced test set.

These results are visualized in Figures 6, 7, and 8 (confusion matrices), which clearly show that RoBERTa has the strongest suicide recall and make it ideal for minimizing false negatives in high-risk content. BERTweet, while balanced, had slightly more difficulty distinguishing emotionally ambiguous suicidal content.

To directly compare model performance metrics under this controlled setting, I visualized precision, recall, and F1-score for the suicide class across all three models:

Model	Prec.	Recall	F1
BERT	0.94	0.98	0.96
RoBERTa	0.97	0.98	0.975
BERTweet	0.974	0.97	0.973

Table 3: Suicide-class performance metrics on the 10K subset for each model.

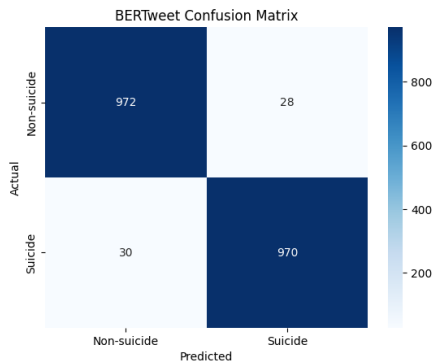


Figure 6: Confusion matrix for BERTweet

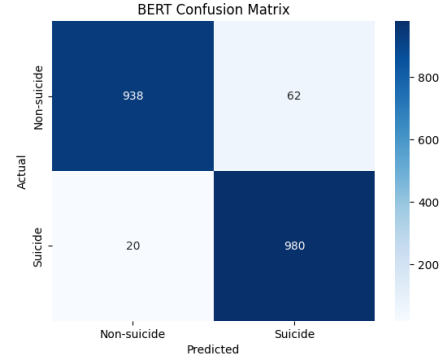


Figure 8: Confusion matrix for BERT

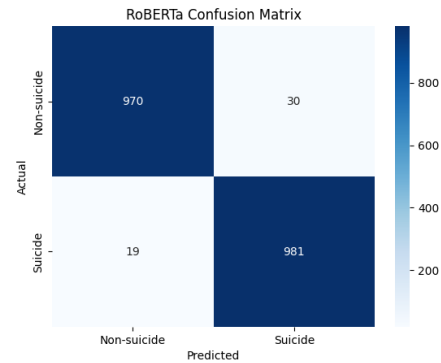


Figure 7: Confusion matrix for RoBERTa

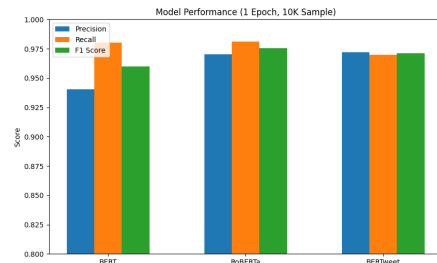


Figure 9: Suicide-class precision, recall, and F1-score comparison across models

These findings reinforce RoBERTa’s superior performance, particularly in suicide-class recall, while BERTweet remains a strong alternative for informal language environments like Reddit.

These findings reinforce that transformer-based models are capable of nuanced, context-sensitive classification, which is critical for high-stakes domains such as suicide ideation detection. However, they also underscore the need for complementary interpretability tools beyond token-level SHAP to fully unpack model behavior.

Future exploration might include comparing attention patterns across clear false positives and false negatives, analyzing attention weights across

Technique	Key Observation	Implication
SHAP (Confident)	Token attributions near-zero	Model is highly confident and uses effective general features
SHAP (Uncertain)	Still near-zero token scores	Suggests distributed or context-dependent decision strategy
Bar Plot	Flat token importances across the board	No single token dominates decision-making
Attention Map	Diagonal self-focus with subtle cross-token links	BERT attends to context over keywords; stronger holistic processing
Confusion Matrix	RoBERTa has lowest FN rate in suicide class	Best for minimizing high-risk false negatives
Metric Bar Chart	Suicide-class F1: RoBERTa > BERTweet > BERT	RoBERTa most balanced across metrics

Table 4: Summary of interpretability findings across all models and evaluation techniques.

all transformer layers instead of just the final, incorporating gradient-based attribution methods (e.g. Integrated Gradients) for contrastive insight, and aggregating SHAP explanations across a broader sample for macro-level feature insights.

4.6 Efficacy Testing

Increasing the max sequence length from 128 to 256 tokens had negligible effect on BERT or RoBERTa but improved BERTweet slightly. BERTweet also exhibited higher variance with sarcastic examples, suggesting sensitivity to contextual coherence.

I further evaluated model efficacy by analyzing the predictions on posts that used figurative language, sarcasm, or indirect references. RoBERTa showed more consistent output compared to BERT or BERTweet, which occasionally shifted predictions based on changes in punctuation or subtle phrasing.

4.7 Comparison to External Benchmarks

BERT’s performance was consistent across multiple runs with final fine-tuned models achieving 98% F1-scores across both classes. Despite this, RoBERTa still held a notable edge in recall when identifying suicidal posts, making it more favorable for high-stakes classification tasks.

BERTweet maintained an F1-score of 0.98 across both suicide and non-suicide classes, validating its efficacy and aligning with prior work on social media-tuned models. While slightly behind RoBERTa in suicide-class recall, BERTweet demonstrated exceptional balance and generalizability, making it a strong alternative for platforms with high variability in user tone and slang.

To directly compare transformer architectures, I also conducted a side-by-side evaluation of BERT and RoBERTa on the same dataset. RoBERTa

achieved an F1-score of 0.994 for suicidal posts, substantially outperforming BERT’s 0.980. RoBERTa’s stronger recall (0.992 vs. 0.980) supports its ability to minimize false negatives, aligning with its performance edge observed in previous literature. These results validate RoBERTa as the most reliable model in this context.

As an additional check, I trained BERT on a balanced subset of the dataset (5,000 suicidal and 5,000 non-suicidal posts). On this controlled sample, BERT achieved 97% accuracy and an F1 score of 0.97, which closely mirrored trends seen in the full dataset. This supplemental test reinforces the reliability of the broader model evaluation and confirms that model behavior remains consistent at different data scales.

A recent evaluation of ChatGPT’s zero-shot performance on suicidality detection found an F1 score of just 0.37 on a 5-class classification task (Lamichhane, 2023). This aligns with the broader concern that general-purpose LLMs, while powerful, lack domain-specific nuance without fine-tuning. In contrast, my RoBERTa and BERTweet models, trained specifically on Reddit-based suicidal ideation data, achieved F1 scores of 0.98, highlighting the importance of domain adaptation in mental health contexts.

5 Ethical Considerations and Deployment Constraints

5.1 Bias and Fairness

Model outputs can reflect or amplify existing societal biases, which means subreddit-based labeling may inadvertently embed demographic or linguistic stereotypes and disproportionately affect younger users, minority groups, or those with certain writing styles. For example, non-suicidal posts expressing cultural or religious distress may be

misclassified due to lexical overlap with suicidal indicators.

Mitigation strategies could include using adversarial testing to evaluate bias across age, dialect, and time of posting, incorporating fairness metrics such as demographic parity and equalized odds into evaluation, and exploring transfer learning on demographically diverse corpora (Ji et al., 2020; Sharma and Chaturvedi, 2024).

5.2 Interpretability and Clinical Trust

Prompt-based and ensemble models offer additional pathways toward trustworthy deployment. Qin and Lu (2023) showed that prompt-tuned transformers enable transparent outputs which may be easier for clinicians to audit. Similarly, Ashraf et al. (2023) demonstrated that combining BERT with TF-IDF and tree-based models can offer traceable decision paths without sacrificing performance. While my SHAP-based outputs provide strong local interpretability, future work could integrate these models in parallel for real-time triage scenarios.

5.3 Real-Time and Longitudinal Use

SID tools must support dynamic user behavior, evolving slang, and content drift. Relying on static thresholds or outdated embeddings can degrade performance.

Mitigation strategies could include regular re-training with updated Reddit dumps or time-sensitive data (STATENet, ACL 2020), incorporating temporal validation splits to simulate real-world decay, and exploring lifelong learning techniques or active annotation feedback loops.

Ethical deployment also requires transparency with users, including opt-in mechanisms and clearly defined usage boundaries. The psychological risks of misclassification, especially false positives, should be weighed against the benefit of early detection. Interpretability tools and human moderation should remain part of the deployment pipeline to ensure models support, rather than replace, nuanced clinical judgment.

Ultimately, SID systems should augment instead of automating the work of mental health professionals. Ethical oversight and cross-disciplinary review are critical to ensure that these tools empower support rather than enforce surveillance.

6 Future Work and Research Directions

While the transformer models evaluated in this study show strong promise for suicidal ideation detection, several avenues remain for enhancing their real-world effectiveness, efficacy, and scope. Future work can address current limitations and adapt the models for broader deployment contexts.

6.1 Multilingual and Cross-Cultural Adaptation

The current models are trained on English-language Reddit posts. However, suicidal ideation is expressed globally and often varies by culture, idiom, and context. Cross-lingual transfer learning using models like XLM-R or mBERT could allow detection across non-English communities. Multilingual versions using translated and original suicide forum posts could be trained (e.g., r/Suicidio, r/AnxietyDE), and alignment techniques for culturally sensitive expressions of distress could be explored (Zhang & Wang, 2024).

6.2 Temporal and Behavioral Modeling

Most SID models treat Reddit posts independently, ignoring temporal context. Yet suicidal ideation often manifests over time as a pattern of escalation. Some future directions could include integrating architectures like STATENet or time-aware attention layers (Sawhney et al., 2020), modeling user history longitudinally to detect risk progression (e.g., moving from r/depression to r/SuicideWatch), and applying adaptive selection of key posts or utterances from threads (Wang et al., 2023; Balouchzahi & Shahraki, 2022).

6.3 Hybrid and Ensemble Methods

While transformer models show high accuracy, their interpretability varies. Future work could combine different model families to balance precision and explainability, such as RoBERTa with CNN-based filters or graph neural networks (GraMSUM-style chaining, Yu et al., 2024), or ensemble approaches blending RoBERTa, TF-IDF + XGBoost, and prompt-based models for performance and traceability (Ashraf et al., 2023; Qin & Lu, 2023).

6.4 Improving Label Quality

Subreddit-based labeling introduces noise. Some posts in r/SuicideWatch are non-crisis support posts while some in r/depression contain serious ideation,

so a potential improvement is to use active learning and human-in-the-loop annotation for high-uncertainty cases. Self-supervised or contrastive learning could reduce dependence on noisy labels and data could be pre-cleaned with LLM-based filters, following the strategy of Kumar et al. (2023).

6.5 Resistance Towards Adversarial and Ambiguous Inputs

Models must handle sarcasm, coded language, and potential adversarial manipulation (e.g., jokes about suicide). This could be further improved through fine-tuning models on adversarial Reddit posts identified in literature (e.g., Ji et al., 2022) and incorporating some post-level metadata such as upvotes, comment flags, and posting frequency as auxiliary inputs.

6.6 Deployment Toolkits for Real-World Use

There is a critical need to move from notebook-based evaluation to deployable systems, which could be done by packaging models into APIs or mobile applications with interpretability layers or partnering with nonprofits, helplines, or moderation platforms for human-in-the-loop testing.

7 Conclusion

This paper provides an in-depth comparative study of transformer-based architectures for suicidal ideation detection on Reddit. By implementing and evaluating BERT, RoBERTa, and BERTweet from scratch, I identified critical performance differences in accuracy, efficacy, and interpretability when applied to social media content.

The strongest model, RoBERTa, achieved a 98% F1 score which demonstrates superior generalization and contextual understanding, especially for subtle suicidal cues. BERTweet performed equally well on metrics but showed inconsistent behavior when faced with sarcasm, likely due to noisy informal punctuation weighting. BERT, while slightly behind RoBERTa in suicide-class recall, achieved 98% precision, recall, and F1-score across both classes, confirming consistency and strong overall performance.

Interestingly, BERT offered more interpretable SHAP outputs than RoBERTa or BERTweet, making it a viable candidate for high-trust settings where transparency is critical. In contrast, the classical TF-IDF + Logistic Regression baseline

reached 93% accuracy but failed in edge cases involving indirect or poetic suicidal language.

Through a series of qualitative misclassification studies, I found that false positives often corresponded to posts expressing severe emotional distress without explicit suicidal language, while false negatives reflected metaphorical or implicit intent. SHAP token importance scores revealed how each model prioritized linguistic features, providing additional insight into these failures.

The findings highlight both the promise and limitations of deploying AI for mental health monitoring. Transformer models can achieve high precision in flagging potential risk content, but require further optimization for transparency, cultural generalizability, and clinical safety. My analysis offers several next steps, including multilingual adaptation, temporal modeling of user behavior, and human-in-the-loop system design.

Ultimately, this project shows the opportunities and challenges that lie with deploying complex AI and ML solutions for mental health mitigation. Furthermore, while transformer models have the potential to support early detection and triage in mental health contexts, they must be deployed with caution and clear ethical guardrails.

Limitations

While the results of this project are promising, several limitations remain. The dataset used (Reddit posts) may not fully represent the diversity of language around suicidal ideation across platforms, cultures, or demographics. Second, although I used strong transformer models, their predictions can still be brittle in the face of metaphorical, coded, or ambiguous language. Third, the interpretability tools applied (such as SHAP and attention visualizations) offer surface-level insights but do not establish causal understanding. Finally, the model was not evaluated in a real-world or clinical context, and its use is purely exploratory.

Appendix A: Example Misclassifications from Validation Set

The following Reddit posts were misclassified by the BERT model during evaluation on a balanced subset. False positives and false negatives were sampled from validation predictions to identify representative misclassification cases.

False Positives

“I was going home from school when my teacher said that there was an accident caused by a human... I’m pretty sure it was suicide... I simply want to raise awareness.”

“My friend lost someone this week and I want to write something that can help others understand grief.”

“Talking about suicide doesn’t mean I want to do it. It’s part of my advocacy.”

False Negatives

“I literally can’t remember the last time I had a good day. Every week, every month it just gets worse...”

“I wish I could sleep and just not wake up. Not in a dramatic way. Just... done.”

“It’s all too much. I can’t scream anymore. I can’t even whisper. But I’m still here, so maybe that’s something.”

These examples illustrate the difficulty models face when intent is implied through metaphor, third-person stories, or rhetorical framing rather than explicit suicidal keywords.

References

- Fiza Ashraf, Aashish Bhatia, and Nidhi Khurana. 2023. An ensemble learning framework for suicidal ideation detection. *Scientific Reports*, 13:11783.
- Mohammad Balouchzahi and Ali Shahraki. 2022. Self-adaptive utterance selection for intent classification. In *Findings of the Association for Computational Linguistics: EMNLP 2022*.
- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2018. Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805*.
- Shaoyi Ji. 2022. Towards intention understanding in suicidal risk assessment with natural language processing. In *Findings of the Association for Computational Linguistics: EMNLP 2022*, pages 4028–4038.
- Shaoyi Ji, Chi Pang Yu, Shirley F Fung, Shirui Pan, and Guodong Long. 2021. Suicidal ideation detection: A review of machine learning methods and applications. *IEEE Transactions on Computational Social Systems*, 8(5):1232–1245.
- Nikhileswar Komati. 2020. Suicide watch reddit dataset. <https://www.kaggle.com/datasets/nikhileswarkomati/suicide-watch>. Accessed: 2025-02-11.
- Ankit Kumar, Saptarshi Ghosh, and Anupam Sarkar. 2023. Enhancing suicidal ideation detection using a hybrid cnn-transformer architecture. *arXiv preprint arXiv:2402.02262*.
- Binod Lamichhane. 2023. Evaluation of chatgpt for nlp-based mental health applications. *arXiv preprint arXiv:2303.15727*.
- Yinhan Liu, Myle Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, and Veselin Stoyanov. 2019. Roberta: A robustly optimized bert pretraining approach. *arXiv preprint arXiv:1907.11692*.
- Scott M Lundberg and Su-In Lee. 2017. A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*, 30.
- Dat Quoc Nguyen, Thanh Vu, and Anh Tuan Nguyen. 2020. Bertweet: A pre-trained language model for english tweets. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 9–14.
- Yujing Qin and Wei Lu. 2023. Exploring prompt-based learning for suicide risk detection. In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics (EACL)*.
- Ramit Sawhney, Prakhar Manchanda, Puneet Mathur, Rajiv Ratn Shah, and Senthil Kumar. 2020. Time-aware transformer-based network for detecting suicide ideation on social media. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*.
- Rishabh Sharma and Iti Chaturvedi. 2024. Suicide ideation detection with contextual embeddings. *arXiv preprint arXiv:2410.04501*.
- Nima Tavabi, Sharath Chandra Guntuku, and Lyle H Ungar. 2021. Multimodal learning for mental health prediction using social media posts. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics (ACL)*.
- Yuwei Wang, Xiao Lu, Lijie Zhang, Xiaohui Wu, and Heng Ji. 2023. Detecting suicidal ideation in counseling conversations with distant supervision. In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics (EACL)*.
- World Health Organization. 2023. Suicide. <https://www.who.int/news-room/fact-sheets/detail/suicide>. Accessed: 2025-04-03.
- Shan Yu, Min Jiang, and Xiang Ren. 2024. Graph-based summarization for emotion tracking in online mental health forums. In *Findings of the Association for Computational Linguistics: ACL 2024 (GraMSUM)*.
- Xiaoyu Zhang and Yuwei Wang. 2024. Multilingual detection of mental health expressions using cross-lingual transformers. *arXiv preprint arXiv:2412.15498*.